Qn. Set Code-1

Semester: 6th
Programme: B.Tech
Branch: CSE, IT, CSCE, CSSE

SPRING END SEMESTER EXAMINATION-2023

6th Semester B.Tech

CLOUD COMPUTING

IT 3022

(For 2021 (L.E), 2020 & Previous Admitted Batches)

Time: 3 Hours | Full Marks: 50

Answer any SIX questions.
Question paper consists of four SECTIONS i.e. A, B, C and D.
Section A is compulsory.
Attempt minimum one question each from Sections B, C, D.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Briefly describe the early technologies that influenced the development of cloud-based businesses.

(b) What is the main benefit of using mobile computing in cloud environments?

(c) Briefly describe one popular Infrastructure-as-a-Service (IaaS) provider other than Amazon Web Services (AWS).

(d) Briefly describe the architecture of cloud on the basis of virtual switches load balancing.

(e) Briefly describe the EC2 feature in AWS.

(f) What are the main benefits of using load balancing in cloud environments?

(g) Give an example of a Software-as-a-Service (SaaS) application that you use regularly.

(h) Briefly describe one popular hypervisor used in virtualization.

(i) Briefly describe the cloud issues and challenges faced in e-Governance.

(j) What is VM Provisioning? Briefly describe its process.

SECTION-B

2. (a) Discuss the key differences among private, public, hybrid, and community cloud models. Provide examples of situations where each would be appropriate. [4]

(b) In recent years, many organizations have migrated their IT infrastructure to the cloud. Analyze the factors that have led to this trend, and discuss the benefits and challenges associated with this shift. [4]

3. (a) Examine the fundamental concepts of distributed systems and grid computing. Which one do you think is the most suitable for cloud computing, and why? [4]

(b) Describe the advantages and disadvantages of using cluster and mobile computing for cloud-based systems. What are the key challenges that must be overcome to ensure their effectiveness? [4]

SECTION-C

4. (a) Contrast the concept of virtual machine migration, and discuss how it can be used to improve the availability and reliability of cloud-based systems. [4]

(b) Compare and contrast different types of hypervisors with examples and their role in cloud computing. [4]

5. (a) What is virtualization? Describe para and full virtualization architectures. Compare and contrast between both. How does virtualization enable cloud-based systems to be more flexible and scalable. [4]

(b) Compare and contrast the features and capabilities of AWS, Google Cloud Platform (GCP), and Microsoft Azure, including their strengths and weaknesses in different application scenarios. [4]

6. (a) Analyze the importance of load balancing and horizontal & vertical scaling in resource management. What are the key challenges associated with these techniques, and how can these be overcome? [4]

(b) Analyze the scheduling issues in cloud computing, and explain the different types of job and task scheduling. How can scheduling be used to improve resource utilization in the cloud? [4]

SECTION-D

7. (a) Investigate the challenges and opportunities associated with cloud computing in the context of mobile devices and wireless sensor networks. [4]

(b) Summarize the challenges and issues associated with cloud security and privacy, and describe the measures that can be taken to mitigate these risks. [4]

8. (a) Analyze the benefits and challenges associated in cloud computing applications in different domains. [4]

(b) Illustrate the concept of wireless sensor networks. Analyze how they can be integrated with cloud computing and discuss the benefits. [4]

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